Questions- Group I

1. (8 p) $f(x) = x^3 - x - 1$ fonksiyonunu $[1, 2]$ aralığı üzerinde ele alınız. İkiye bölme metodunu kullanarak $[1, 2]$ aralığı üzerinde f için p_3 ü bulunuz.

(*Use the Bisection method to find p_3 for the function $f(x) = x^3 - x - 1$ on $[1, 2]$.*)

A) $\frac{11}{2}$ B) $\frac{11}{4}$ C) $\frac{11}{32}$ D) $\frac{11}{16}$ E) $\frac{11}{8}$

$f(1) = -1 < 0 \quad f(2) = 5 > 0$

$a_1 = 1 \quad b_1 = 2 \quad p_1 = \frac{a_1 + b_1}{2} = \frac{3}{2} \quad f(p_1) = f(\frac{3}{2}) = \frac{7}{8} > 0$

$f(1) \cdot f(p_1) < 0 \quad \therefore a_2 = 1 \quad b_2 = 3/2 \quad p_2 = \frac{a_2 + b_2}{2} = \frac{5}{4}$

$f(p_2) = f(\frac{5}{4}) = -\frac{19}{64} \quad f(b_2) \cdot f(p_2) < 0$

$\therefore a_3 = p_2 = \frac{5}{4}, \quad b_3 = b_2 = \frac{3}{2} \quad \therefore p_3 = \frac{a_3 + b_3}{2} = \frac{11}{8}$

2. (8 p) $g(x) = 3^{-x}$ olarak verilsin. $p_0 = 0$ olmak üzere sabit nokta iterasyon yöntemini kullanarak p_3 ü bulunuz.

(*Use a fixed point iteration method to find p_3 for the function $g(x) = 3^{-x}$ on $[1, 2]$. Use $p_0 = 0$.*)

A) $3^{-1/3}$ B) $3^{-1/6}$ C) $3^{-1/9}$ D) $3^{-1/12}$ E) $3^{-1/15}$

$p_n = g(p_{n-1}), \quad n = 1, 2, \ldots$

$p_1 = g(p_0) = g(0) = 3^{-0} = 1$

$p_2 = g(p_1) = g(1) = 3^{-1} = 1/3$

$p_3 = g(p_2) = g(\frac{1}{3}) = 3^{-1/3}$

3. (10 p) $f(x) = xe^{-x}$ olarak verilsin. Newton-Raphson (Newton) formülü ile $p_0 = 0.5$ ten başlayarak p_2 değerini bulunuz.

(*Use a Newton-Raphson (Newton's) method to find p_2 for the function $f(x) = xe^{-x}$ with $p_0 = 0.5$.*)

A) $\frac{-1}{6}$ B) $\frac{1}{6}$ C) $\frac{-1}{3}$ D) $\frac{1}{3}$ E) $\frac{-1}{12}$

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} = \frac{p_{n-1}^2}{p_{n-1} - 1}$$

$$\left\{ f'(x) = (1-x)e^{-x}, \quad p_n = p_{n-1} - \frac{p_{n-1}\, e^{-p_{n-1}}}{e^{-p_{n-1}}(1 - p_{n-1})} \right\}$$

$$k=1 \Rightarrow p_1 = \frac{p_0^2}{p_0 - 1} = \frac{1/4}{1/2 - 1} = -1/2$$

$$k=2 \Rightarrow p_2 = \frac{p_1^2}{p_1 - 1} = \frac{1/4}{-1/2 - 1} = -1/6.$$

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4. (12 p)

Tabloda verilen değerlerle Newton bölünmüş farklar polinomunu kullanarak $f(0.5)$'in yaklaşık değerini hesaplayın.

(*Find the approximate value of $f(0.5)$ using Newton's divided difference polynomial with the following data.*)

x	-2	-1	0	1
f(x)	-1	3	1	-1

A) 0.625 B) -0.428 C) -0.375 D) 0.333 E) -0.625

Solutions :

(4) $P_3(x) = f[x_0] + f[x_0,x_1](x-x_0) + f[x_0,x_1,x_2](x-x_0)(x-x_1)$

$+ f[x_0,x_1,x_2,x_3](x-x_0)(x-x_1)(x-x_2)$

$x_0 = -2, \quad x_1 = -1, \quad x_2 = 0, \quad x_3 = 1$

$$f[x_0,x_1] = \frac{f[x_1]-f[x_0]}{x_1-x_0} = \frac{3-(-1)}{-1-(-2)} = 4$$

$$f[x_0,x_1,x_2] = \frac{f[x_1,x_2]-f[x_0,x_1]}{x_2-x_0} = \frac{-2-4}{1-(-1)} = \frac{-6}{2} = -3$$

$$f[x_1,x_2] = \frac{f[x_2]-f[x_1]}{x_2-x_1} = \frac{1-3}{0-(-1)} = -2$$

$$f[x_0,x_1,x_2,x_3] = \frac{f[x_1,x_2,x_3]-f[x_0,x_1,x_2]}{x_3-x_0} = \frac{0-(-3)}{1-(-2)} = 1$$

$$f[x_1,x_2,x_3] = \frac{f[x_2,x_3]-f[x_1,x_2]}{x_3-x_1} = \frac{-2-(-2)}{-1-3} = 0$$

$$f[x_2,x_3] = \frac{f[x_3]-f[x_2]}{x_3-x_2} = \frac{-1-1}{1-0} = -2$$

$P_3(x) = -1 + 4(x+2) - 3(x+2)(x+1) + 1(x+2)(x+1)x$

$$P_3(x) = x^3 - 3x + 1$$

$$f(0.5) \approx P_3(0.5) = \left(\frac{1}{2}\right)^3 - \frac{3}{2} + 1 = \frac{1}{8} - \frac{1}{2} = -\frac{3}{8} = -0.375$$

A) 0.625
B) -0.428
C) -0.375
D) 0.333
E) -0.625

A] - 0.375

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(5) $f(x)=\cos(\pi x)$, $x_0=0$, $x_1=\frac{1}{2}$ $(n=2)$

a) Degree of the Hermite polynomial is $2n+1=3$:

$H_3(x)=a_0+a_1x+a_2x^2+a_3x^3$

$H_3'(x)=a_1+2a_2x+3a_3x^2$, $f'(x)=-\pi\sin(\pi x)$

$H_3(x_0)=f(x_0)\Rightarrow H_3(0)=\cos(0)\Rightarrow \boxed{a_0=1}$

$H_3(x_1)=f(x_1)\Rightarrow H_3(\frac{1}{2})=\cos(\frac{\pi}{2})\Rightarrow a_0+\frac{a_1}{2}+\frac{a_2}{4}+\frac{a_3}{8}=0$ (1)

$H_3'(x_0)=f'(x_0)\Rightarrow H_3'(0)=-\pi\sin(0)\Rightarrow \boxed{a_1=0}$

$H_3'(x_1)=f'(x_1)\Rightarrow H_3'(\frac{1}{2})=-\pi\sin(\frac{\pi}{2})\Rightarrow a_1+2a_2\frac{1}{2}+3a_3\frac{1}{4}=-\pi$ (2)

(1); (2) : $\begin{cases} 2a_2+a_3=-8 \\ 4a_2+3a_3=-4\pi \end{cases}$

Solving this system we have

$\boxed{a_3=-4\pi+16}$

$\Rightarrow a_2=\frac{1}{2}(-8-a_3)=-4-(-2\pi+8)\Rightarrow \boxed{a_2=2\pi-12}$

Hence

$\boxed{H_3(x)=1+(2\pi-12)x^2+(-4\pi+16)x^3}$

b) $\cos(\frac{\pi}{4})\approx H_3(\frac{1}{4})=1+(2.3-12)\frac{1}{16}+(16-4.3)\frac{1}{64}$

$=1-\frac{6}{16}+\frac{1}{16}=1-\frac{5}{16}=\frac{11}{16}$

$H_3(\frac{1}{4})\approx 0.6875$ (Exact value $\cos(\frac{\pi}{4})\approx 0.7071$)

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(6)

$$S(x) = \begin{cases} S_0(x) = 1 + Bx + 2x^2 - 2x^3 & , \ 0 \le x < 1 \\ S_1(x) = 1 + b(x-1) - 4(x-1)^2 + 7(x-1)^3 & , \ 1 \le x \le 2 \end{cases}$$

$x_0 = 0, \ x_1 = 1, \ x_2 = 2$

i) $S_0(x_0) = f(x_0) \Rightarrow S_0(0) = f(0) \Rightarrow 1 = f(0)$

$S_0(1) = S_1(1) \Rightarrow 1 + B = 1 \Rightarrow \boxed{B = 0}$

$S_1(2) = f(2) \Rightarrow 1 + b - 4 + 7 = f(2) \Rightarrow b + 4 = f(2)$

ii) $S_0'(x_1) = S_1'(x_1) \rightarrow S_0'(1) = S_1'(1)$;

$\Rightarrow B + 4 - 6 = b \Rightarrow \boxed{b = -2}$

$S_0'(x) = B + 4x - 6x^2$

$S_1'(x) = b - 8(x-1) + 21(x-1)^2$

$S_0''(x) = 4 - 12x$

$S_1''(x) = -8 + 42(x-1)$

iii) $S_0''(x_1) = S_1''(x_1) \rightarrow S_0''(1) = S_1''(1)$

$\Rightarrow 4 - 12 = -8$ ✓✓

iv) Clamped BCs: $S_0'(x_0) = f'(x_0), \ S_1'(x_2) = f'(x_2)$

$f'(0) = S_0'(0) \Rightarrow \boxed{f'(0) = B = 0}$

$f'(2) = S_1'(2) = b - 8 + 21 = -2 - 8 + 21 \Rightarrow \boxed{f'(2) = 11}$

A) 3

B) 5

C) 0

D) 11

E) 9

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7. **(10 p)** Birinci mertebeden türev için iki-nokta formüllerini kullanarak aşağıda verilen tablodaki a ve b değerlerini belirleyiniz.

(Use proper two-point difference formula for the first-order derivative to determine the values a and b in the following table.)

x	$f(x)$	$f'(x)$
0.5	1.9	a
0.6	b	2.2

A) $a = 2.12, b = 2.02$
B) $a = 2.12, b = 2.2$
C) $a = 2.2, b = 2.02$
D) $a = 2.02, b = 2.12$
(E) $a = 2.2, b = 2.12$

From Backward-Difference Formula:

$$f'(0.6) = \frac{f(0.6) - f(0.5)}{0.1} \Rightarrow 2.2 = \frac{f(0.6) - 1.9}{0.1} \Rightarrow \boxed{f(0.6) = 2.12 = b}$$

From Forward-Difference Formula:

$$f'(0.5) = \frac{f(0.6) - f(0.5)}{0.1} \Rightarrow f'(0.5) = \frac{2.12 - 1.9}{0.1} \Rightarrow \boxed{f'(0.5) = 2.2 = a}$$

8. **(8 p)** $f(0) = 8$, $f(1) = 5$, $f(2) = 3$, $f(3) = 2$ ve $f(4) = 3$ olacak şekilde verilmiş ikinci mertebeye kadar sürekli türevlere sahip $f(x)$ fonksiyonu için $f''(2)$ türevini ara nokta formülünü kullanarak $h = 2$ adımıyla yaklaşık olarak hesaplayınız.

(Given a twice continuos differentiable function $f(x)$ for which $f(0) = 8$, $f(1) = 5$, $f(2) = 3$, $f(3) = 2$ and $f(4) = 3$. Use the second derivative midpoint formula with the step $h = 2$ to approximate $f''(2)$.)

(A) 1.25 B) 1.5 C) 2.5 D) 1 E) 2.25

$$f''(2) = \frac{f(4) - 2f(2) + f(0)}{4} = \frac{3 - 2.3 + 8}{4} = 1.25 \Rightarrow \boxed{f''(2) = 1.25}$$

(Midpoint Formula : $f''(x_0) = \frac{f(x_0+h) - 2f(x_0) + f(x_0-h)}{h^2}$)

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Q **9.** Simpson kuralı kullanarak aşağıdaki integrale yaklaşık değeri hesaplayınız (*Compute the Simpson's rule approximation for the following integral*)

$$\int_0^2 \left(2x\cos(\pi x) - (x-2)^2\right) dx.$$

Çözüm: Simpson formülüne göre

$$\int_a^b f(x)dx \approx \frac{h}{3}\left[f(a) + 4f(\frac{a+b}{2}) + f(b)\right]$$

$a = 0$ ve $b = 2$ olduğuna göre $\frac{a+b}{2} = 1$ oluyor ve $h = 1$ dir. Yerine yazırsak

$$\int_0^2 \left(2x\cos(\pi x) - (x-2)^2\right) dx = \frac{1}{3}\left[-4 + 4 * (-3) + 4\right] = -4$$

olur.

Q **10.** Aşağıdaki integral verilsin (*Consider the integral*)

$$\int_0^4 x^2 e^{3(x-2)^2} dx.$$

$n = 2$ durumunda Gauss kareleme yöntemi uygulayarak bu integrale yaklaşık değeri bulunuz (*Approximate this integral using Gaussian quadrature method with $n = 2$*)

Çözüm:

$$\int_a^b f(x)dx = \int_{-1}^1 \frac{b-a}{2} f(\frac{(b-a)\,t + b + a}{2})dt$$

formülü gereği

$$\int_0^4 x^2 e^{3(x-2)^2} dx = 8\int_{-1}^1 (t+1)^2 e^{12t^2} dt$$

ve n=2 durumunda Gauss kareleme yöntemine göre

$$= 8\left(\left(1 + \frac{1}{\sqrt{3}}\right)^2 + \left(1 - \frac{1}{\sqrt{3}}\right)^2\right) e^4 = \frac{64}{3}e^4$$